Moez A. AbdelGawad

Moez Adel AbdelAziz AbdelGawad Khalil Turk

Assistant Professor — Quantum Informatics and Programming Languages

Informatics Research Institute (IRI), SRTA-City · New Borg ElArab, Alexandria, Egypt

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APPOINTMENTS

- **Assistant Professor**, Quantum Informatics and Programming Languages — Informatics Research Institute (IRI), SRTA-City, Egypt (2013–present).
- **Visiting Scientist** (remote), Computer Science Department, Rice University, Houston TX, USA (2020–present).
- **Adjunct Assistant Professor**, Computer and Systems Engineering Department (CSED), Faculty of Engineering, Alexandria University, Egypt (2009–present; On/Off).

EDUCATION

- Ph.D., Computer Science, **Rice University**, Houston TX, USA — 2011.

  Thesis: NOOP: A Mathematical Model of Object-Oriented Programming (OOP type systems).
- M.Sc., Computer Science, **Alexandria University**, Egypt.

  Thesis: 2D Encryption Mode (2DEM) — a novel encryption mode (security algorithms).
- B.Sc., Computer Science, **Alexandria University**, Egypt.

  Graduation project: Human Face Recognition (digital image processing, neural networks).

RESEARCH INTERESTS

- Primary: Quantum informatics, quantum algorithms, and quantum programming; categorical quantum theory.
- Related: Programming languages theory; object-oriented type systems; applied category theory.
- Earlier: Cryptography and computer security; digital image processing; neural networks and machine learning.

PEER-REVIEWED PUBLICATIONS

- AbdelGawad, M. A. *Towards a Java Subtyping Operad.* Formal Techniques for Java-like Programs (FTfJP'17), ECOOP'17, Barcelona, Spain, 2017.
- AbdelGawad, M. A., Cartwright, R. *A Domain-Theoretic Model of Nominally-Typed Object-Oriented Programming.* Electronic Notes in Theoretical Computer Science (ENTCS), Vol. 301, pp. 3–19, 2014.
- Abdel-Gawad, M. A., Belal, A. A. *2D-Encryption Mode (2DEM).* Proc. SPIE Mathematics of Data/Image Coding, Compression, and Encryption V, with Applications, Seattle, USA, 2003.
- AbdelGawad, M. A. *Using Category Theory in Modeling Generics in OOP* (extended abstract, poster). Applied Category Theory Confer. (ACT'19), Quantum Group, Dept. of Computer Science, Oxford University, UK, 2019.
- Plus 20+ further research articles, preprints, and technical reports (on arXiv and elsewhere) on OOP generics and Java wildcards, domain theory, the Coq proof assistant, partial graph products, and induction / coinduction / fixed points — full list available upon request or via extended CV.

PREPRINTS & TECHNICAL REPORTS

- AbdelGawad, M. A. *Introducing QAL: the Quantum Algorithms Lab.* EasyChair Preprint 16020, 2026.

- AbdelGawad, M. A. *Towards an Order and Category-Theoretic Model of Java Generics.* EasyChair Preprint 3631, 2020.

BOOKS

- AbdelGawad, M. A. *NOOP: A Nominal Mathematical Model of Object-Oriented Programming.* Scholars' Press, 2013. ISBN 978-3-639-51281-6.

In preparation

- Cartwright, R., Parsons, R., AbdelGawad, M. A. *Domain Theory: An Introduction.* Draft available at arXiv:1605.05858 [cs.PL], 2016.

CURRENT PROJECTS

- **QAL — Quantum Algorithms Lab:** a visual, interactive web application for researching and teaching quantum algorithms. A lightweight public version (QAL Lite) is freely available online.
- **Quantum Informatics Lab (QI Lab):** proposed research lab at IRI, SRTA-City (proposal, Dec. 2025).
- **Vision — First *Quantum* FIFA World Cup:** proposed how Saudi Arabia may use quantum informatics and quantum technologies to organize and operate *FIFA World Cup 2034* (concept proposal, June 2026).
- **Outreach — QIAP channel:** founder and moderator of an English-language channel on quantum informatics, algorithms, and programming.

GRANTS & RESEARCH PROPOSALS

- Modeling Java Generics — PI; accepted for STDF funding, 2015.
- Automatic Software Parallelization — Co-PI (PI: Prof. Ahmed Elmahdy, EJUST); accepted for STDF funding, 2015.
- (Proposed) Using Quantum Computing and Supercomputers to Analyze the Spread of COVID-19 — PI, with multiple Alexandria University faculty and students in medical and engineering sciences, 2020.

TEACHING

Interests

- Quantum informatics, quantum algorithms and programming; advanced OOP; algorithms; linear algebra; discrete mathematics (group theory, set theory, lattice theory); applied category theory; visual/interactive app development; 3D animation and visualization (incl. VR/AR); AI/machine learning; OOP type systems; security algorithms.

Courses

- (Under Preparation) Quantum Algorithms and Quantum Informatics using QAL — training and semester-length courses (2026).
- Quantum Computing Training Programme ("The QC Game") — IRI, SRTA-City, undergraduate (Feb. 2022).
- Quantum Programming — CSE Dept. (CSED), Faculty of Engineering, Alexandria University, graduate (Fall 2019); course material freely available online.
- Advanced Compilers (Automatic Software Parallelization) — Egypt-Japan University of Science & Technology (EJUST), graduate (Spring 2014).
- Supervised the Order Theory in Proof Designer (OTPD) B.Sc. graduation project — CSE Dept., Faculty of Engineering, Alexandria University (2009–2010).

- 30+ invited talks on quantum algorithms, OOP type-system modeling, domain theory, OOP and functional programming (Java and Scala), proof assistants, and encryption modes.

FURTHER RESEARCH EXPERIENCE

- Visiting Research Scholar, Category Theory Group (with Prof. Qingguo Li), College of Mathematics, Hunan University, Changsha, China (2015–2016).
- Summer Research Intern, Programming Languages Group (with Prof. Martin Odersky, chief architect of Scala), École Polytechnique Fédérale de Lausanne (EPFL), Switzerland (2005).

LANGUAGES & TECHNICAL SKILLS

- Natural languages: Arabic (native); English (fluent); French (intermediate); German (basic).
- Primary tools/programming languages: Java, C/C++/C#, JavaScript/TypeScript (D3.js, Three.js, WebXR), Scala, Coq, Haskell, OCaml, Python (NumPy, SciPy, Jupyter), SageMath, Matlab, Mathematica — full list available on request.

PERSONAL

- Born in Jülich, (West) Germany (2 years). Lived in: Sheffield, UK (4 years); Houston, TX, US (7 years); Lausanne, Switzerland (6 months); Changsha, China (1 year).
- Very briefly visited Hong Kong (2015; 5 days), Jordan (2006; in transit), KSA (2006; Hajj), Germany (2005; 3 days), France (2005; 1 day), and Qatar (2013; in transit).